

W

BUS7C2

Finance and Accounting for Business

Prifysgol Wreccsam
Wrexham University

Lecture 10: Financial Technology

Prifysgol Wreccsam
Wrexham University



Learning Objective

At the end of this lecture; students will:

1. Evaluate how FinTech has reshaped financial services and business models, from early innovations to current digital disruptions.
2. Analyse key FinTech areas—payments, Insurtech, Regtech, Wealthtech, and Blockchain—highlighting the role of enabling technologies.
3. Propose strategic solutions to FinTech-related risks, opportunities, and regulatory challenges, balancing innovation with governance.

What is Fintech?

Combining the words ‘**financial**’ and ‘**technology**’, fintech is ***technology-enabled financial innovation***, which is changing the way financial institutions provide – and consumers and businesses use – financial services.

(Bank of England, 2023)



- It describes the ***use of technology to deliver financial services and products*** to consumers.
- Any ***innovation in financial sector to assist how people transact business***, from the invention of digital money including cryptocurrency to double-entry bookkeeping etc....
- Originally referred to computer technology applied to the back office of banks/trading firms

“Old” Fintech

- **Bank** were among the first industries to adopt computers and use them for account processing.....huge cost saving opportunities
- Deploying internet banking, i.e. making their existing business functions accessible via a website



- **FinTech**, then, refer to the technical financial market infrastructure that was running the show: ATM, SWIFT etc.
- A tool for the production, *not the disruption*, of standard banking services

Fintech Today

Examples

- ❑ **Innovative and disruptive developments** by tech firms that are beginning to substitute some of the service of classic players in Finance.

Since internet/mobile internet/smartphone revolution

- ❖ *Technological interventions into personal and commercial finance*
- ❖ *Improve and automate the delivery and use of financial service in the area of investment banks, retail banks, financial education, fundraising and nonprofit*

- stock trading apps and websites,*
- peer to peer loan/lending website,*
- online portfolio management service,*
- all-in-one online personal finance management, budgeting tools, raising money for business startup....*



Digital Disruption : The 1st Wave

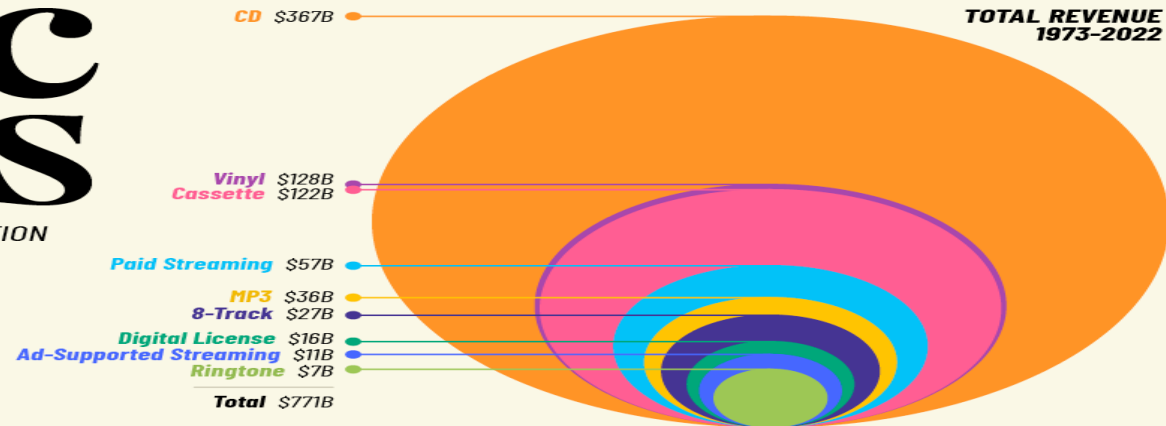
- The vast improvements in information technology and the rapid adoption of the internet soon made a number of business models obsolete.
- Business models *at instant risk of disruption* were those that entailed at their core value proposition:
 - *the delivery of information on physical media, or*
 - *the forwarding or broadcasting of information.*

Digital Disruption : Music Sale Format

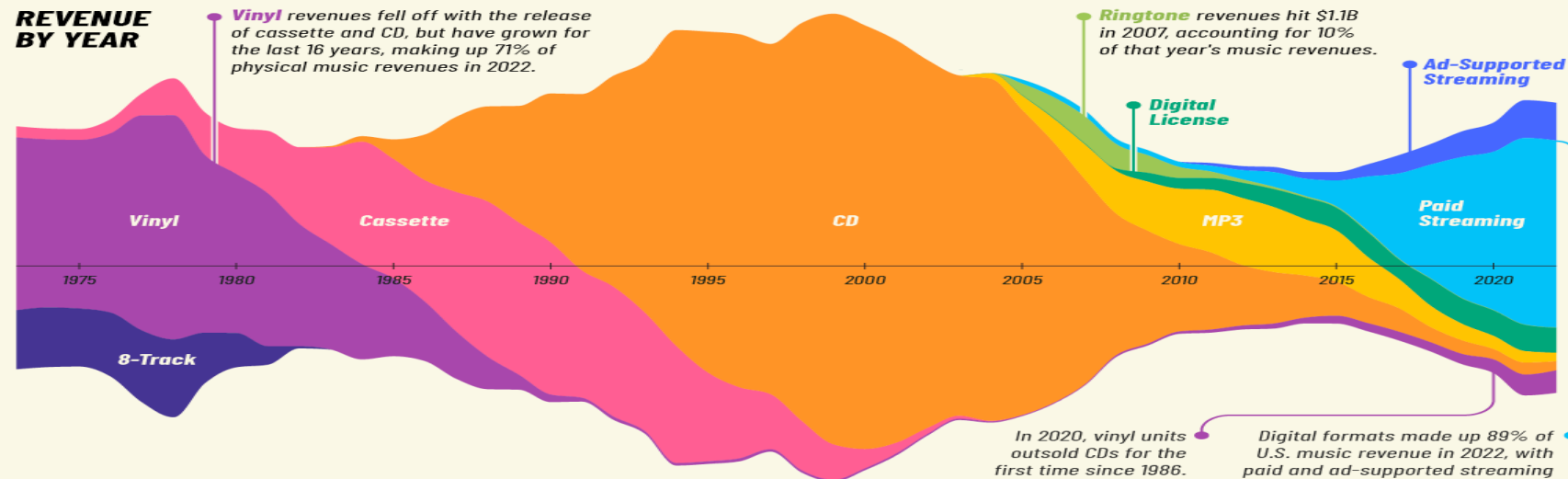
MUSIC SALES

1973 - 2022 ✈ ADJUSTED FOR INFLATION

While digital formats lead the music industry's revenues, vinyl is the only remaining physical format with growing sales.



REVENUE BY YEAR



Source: RIAA Notes: Adjusted for inflation, 2022 USD. Values are at MSRP. Formats with no retail value equivalent are at wholesale price. Cassettes includes reel-to-reel and quadrophonic. Ringtones includes master ringtones, ringbacks, and prior to 2013 music videos, full length downloads, and other mobile. Ad-Supported Streaming includes revenues paid directly for statutory services that are not distributed by SoundExchange and not included in other streaming categories. Paid Streaming includes streaming, tethered, and other paid subscription services not operating under statutory licenses. Digital Licensing includes Sound Exchange Distributions, Synchronizations, and other digital music licensing.

VISUAL CAPITALIST
Visualcapitalist.com

W

Digital Disruption : The 2nd Wave

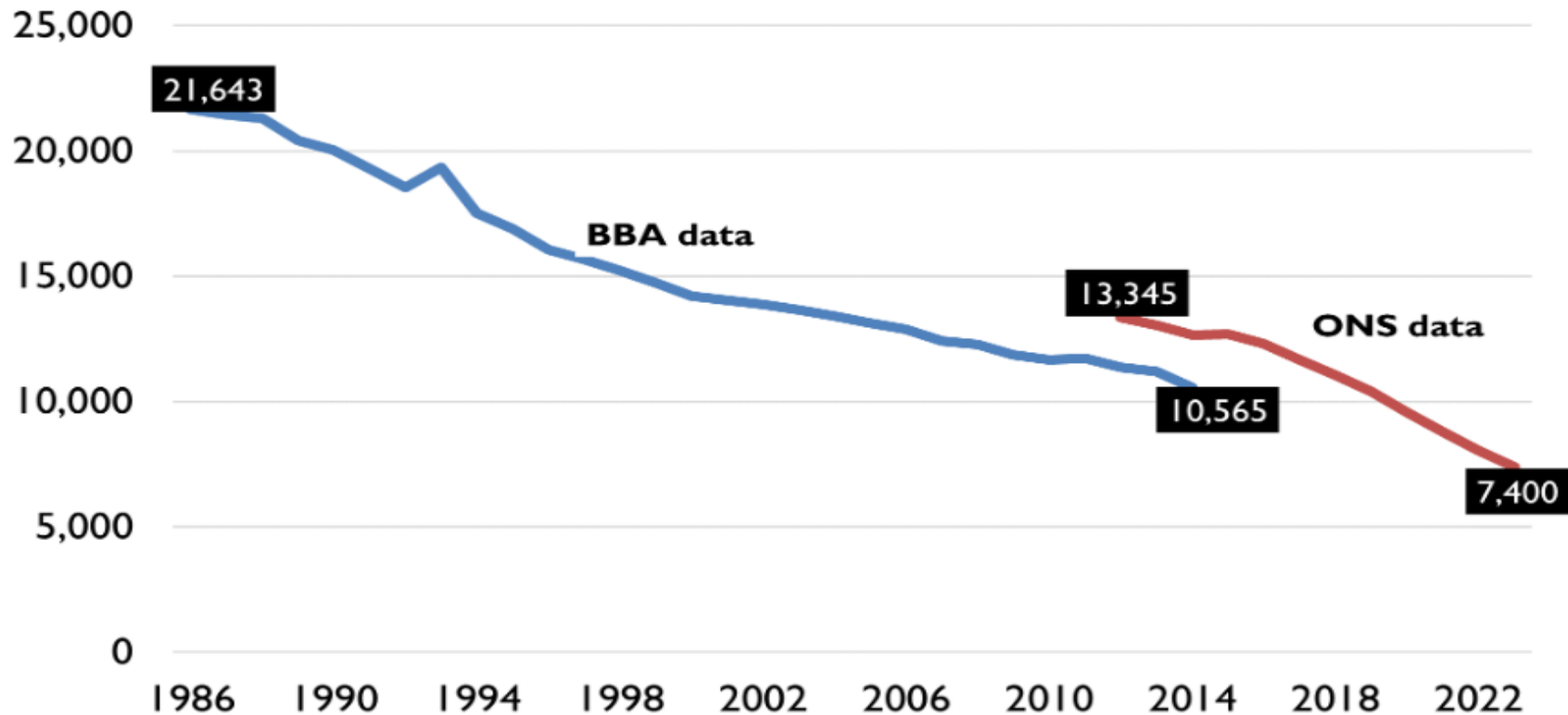
- ***Growing internet user base***, more scope to exploit growing economies of scale.
- Growing **information advantage of tech firms over other players** in the economy.
- Growing **willingness of users to lend their trust** to internet technology if packaged as an app.

Further business models at risk of digital disruption:

- Business models whose value creation depends on **knowing customer characteristics** or enjoying **customer's trust**. In particular: **information-centric** business models of **intermediation**.
- Business models built around the provision of services for which a **new, informed digital intermediary can create substitutes** of better value proposition.

Digital Disruption : *Effect?*

Chart 1. Total number of bank branches and building societies in the UK since 1986



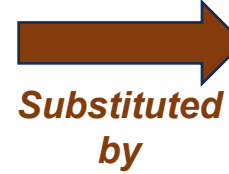
(House of Commons Library, '[Statistics on access to cash, bank branches and ATMs](#)', 1 September 2023; and Office for National Statistics, '[Nomis: Official census and labour market statistics](#)', accessed 16 January 2024)

Digital Disruptors



Airbnb is a perfect example of an informed digital intermediary that brings two parties (host and guest) together and enables mutual trust. As a result, private homes have become viable substitutes for hotel services.

trusted traditional cab driver



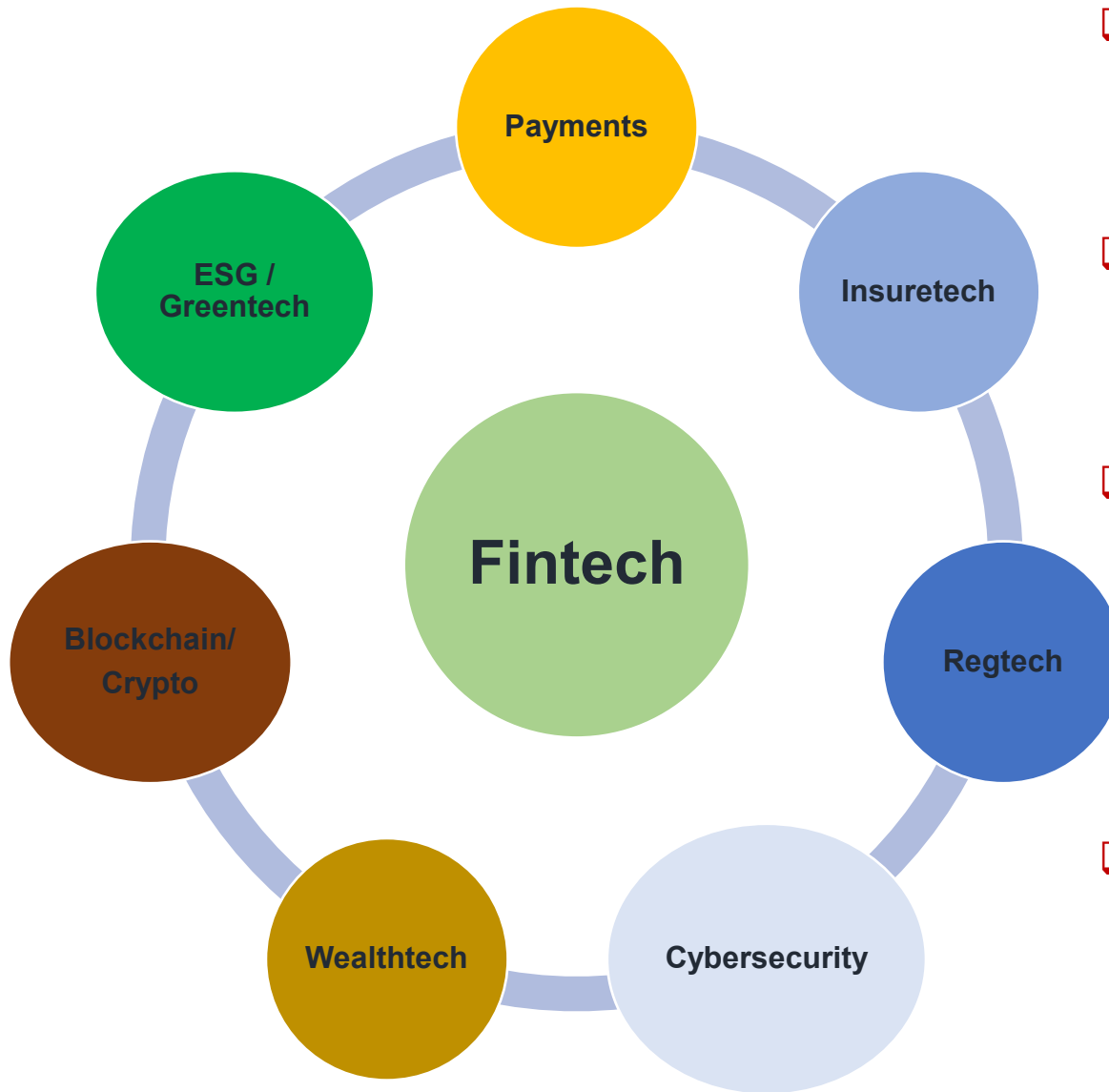
Substituted by

a trusted app



- Traditional cab call centers are information intermediaries.
- **Official cabs = means of trusted transportation**
- **extensive checks and licensing requirements imposed on drivers.**
- Innovation of Uber/Lyft:
 - easier-to-use,
 - better informed and more powerful digital intermediary.

Fintech Segments



❑ Payments

~Digital only banks

~Mobile PoS

❑ Regtech

~Anti-Money Laundry (AML)

~Fraud detection and prevention

❑ Insurtech

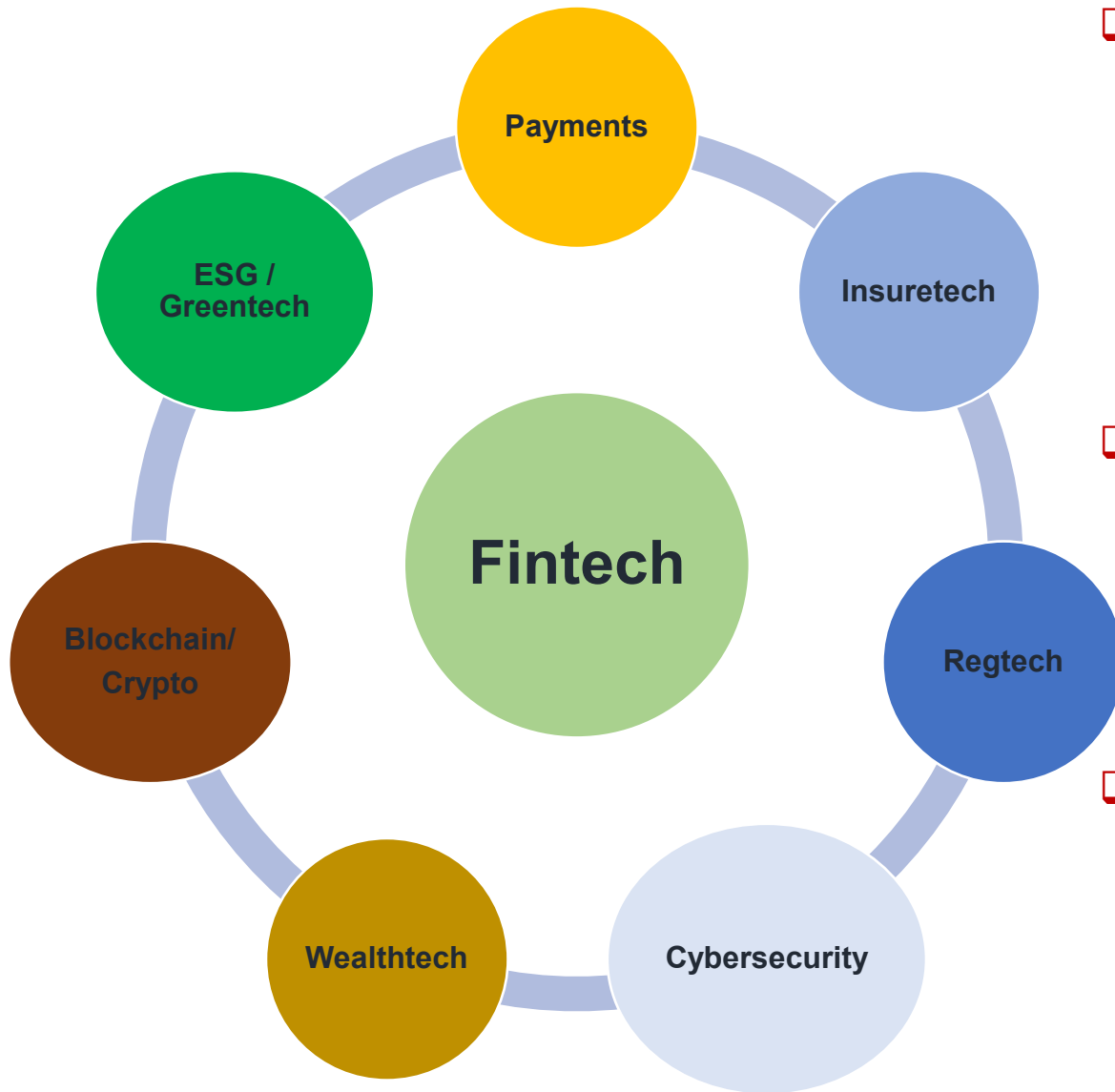
~Personalized Insurance Pricing with IoT and Social Media

~Telematics Insurance - [O2 Smart Vehicle](#)

❑ Cybersecurity

~Digital identity verification – facial and iris recognition

Fintech Segments (2)



❑ **Wealthtech**

~Robo-advisors,
~Portfolio management tools,
~Micro-investment and social trading platforms

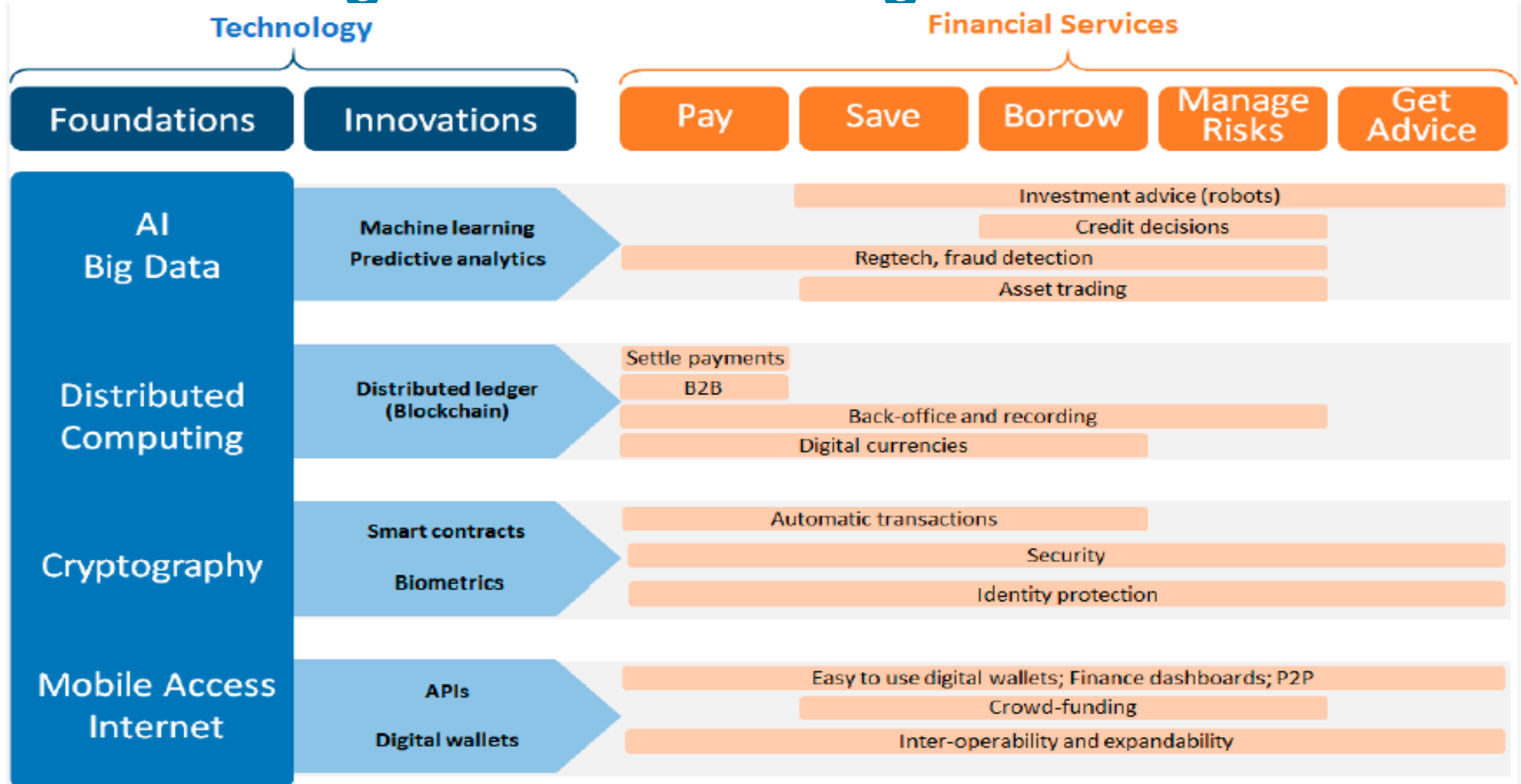
❑ **Blockchain/Crypto**

~Distributed ledger technology (DLT)
~Digital currency
~Non-fungible tokens (NFTs)

❑ **ESG/Greentech**

~Sustainability and fintech – [TreeCard](#);
[Trine P2P solar energy finance](#)

Technologies Transforming Financial Services



Source: IMF staff.

Fintech As Digital Disruptive Innovation

Alternative Payment Rails



Online-based life products



A German digital-only bank



Mass P2P Lending



Robo-Advisors



Mass Crowdfunding



Fintech and Payments

The FinTech Tribes..... Panzarino and Hatami (2021)

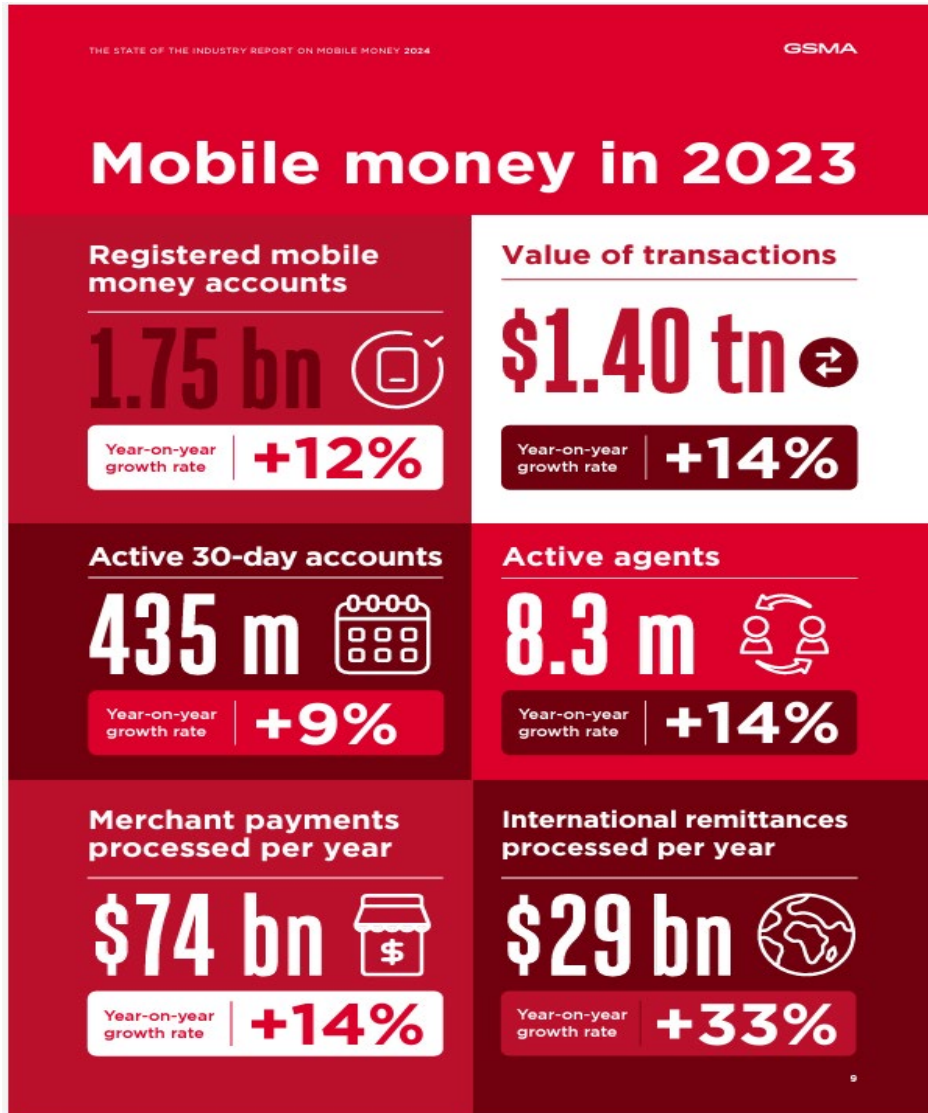
- One tribe is the '***The payments innovators***'.
- US west coast dotcom revolution, industries such as travel, music and retail were upended and bookstores, music retailers and travel agents wiped out. The new entrants needed digital payments.
- **The founding member of the payments innovators was PayPal.**
 - Prior to this, becoming a merchant with traditional providers (i.e. for online card payments) was challenging and P2P payments were practically impossible online.
 - This was a real issue for platforms such as eBay. PayPal was able to grow quickly inside eBay. Other businesses then started to use PayPal.
- **PayPal started a FinTech revolution in the payments system.**

Fintech and Payments (2)

The FinTech Tribes..... Panzarino and Hatami (2021)

- Another tribe is the '***The champion of the unbanked***'.
- Mobile technology applied by this 'tribe' is a formidable force in the democratization of finance.
- **Nubank** (a neobank) in Brazil launched in 2013 to offer a fee-free credit card managed by an app. Targeted Brazil's 60 million unbanked citizens. Now one of the largest challenger banks in the world with over 14 million customers.
- **Paytm** based on mobile phones as e-wallets. Over 400 million active accounts in India and accepted at over 7 million merchants.
- **M-Pesa** (later) successful in Kenya and Tanzania in particular. More recent competitors such as Airtel Money, Equitel, MobiKash.

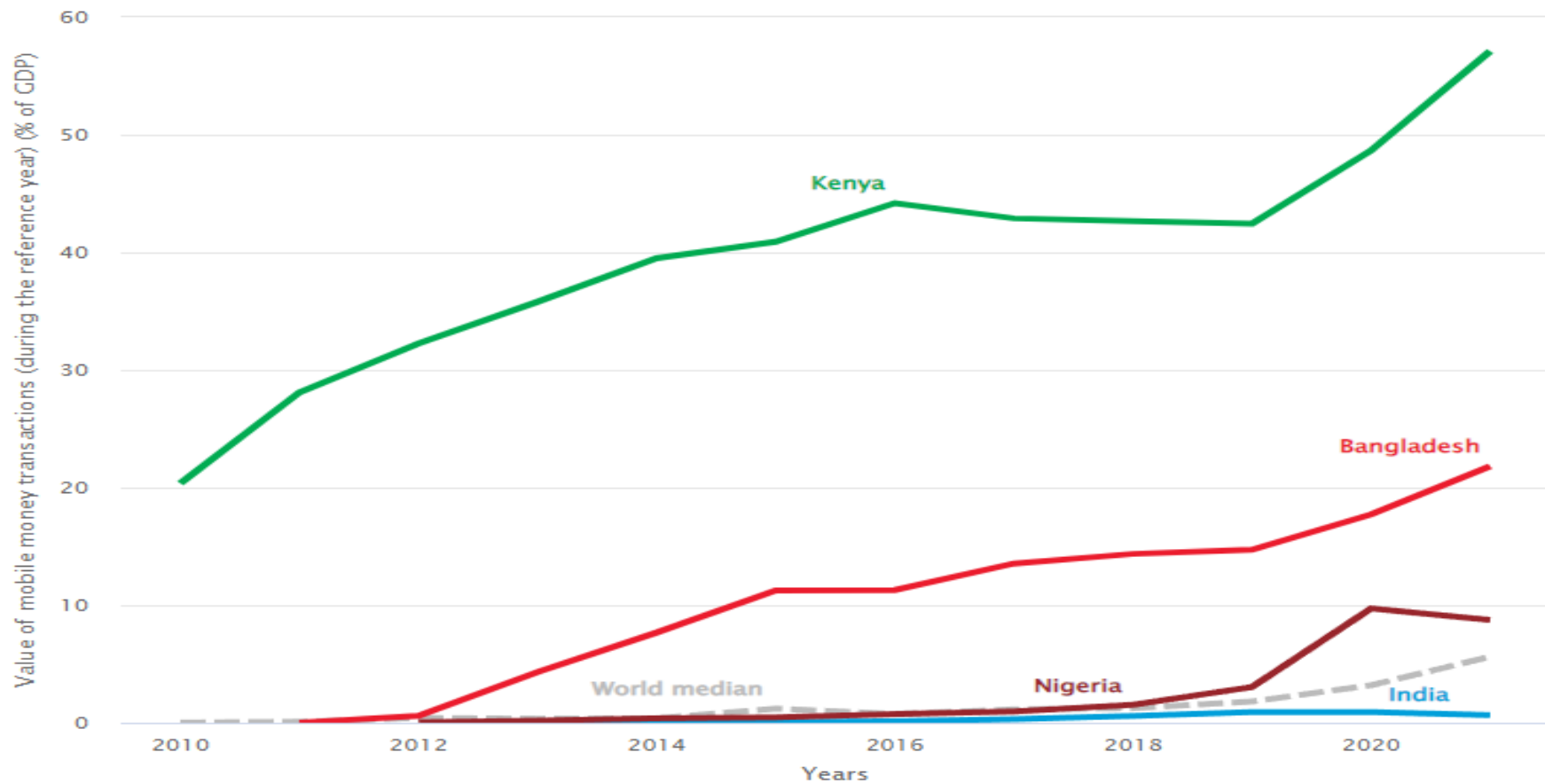
Mobile Banking : Broadening access to financial services



[GSMA 2024](#)

- Increasing global interest of donors, governments, regulators
- Banking and commercial sectors pursuing the use of mobile technologies to deliver financial services in emerging markets
- Leapfrog traditional ‘bricks and mortar’ banking i.e., branches
- Good mobile network infrastructure
- Lower costs associated with development, provision and use of services.
- Additive or transformative roles of mobile banking

Value of mobile money transactions (% of GDP)



Data Source: Financial Access Survey (FAS) | [Download Dataset](#)

For Regional and Income Group breakdowns, visit: [World Bank Country and Lending Groups](#).

Mobile Money : Reflections and Lessons

- a. Lack of retail banking infrastructure provided an opportunity for the service to flourish
- b. Lack of regulatory restrictions or late introduction of elementary regulations allowed for rapid expansion
- c. Need for network externalities i.e.success grows if the network grows
- d. Simplicity of the technology
- e. Improved financial inclusion has huge positive knock-on effects

Digital Currencies : Crypto

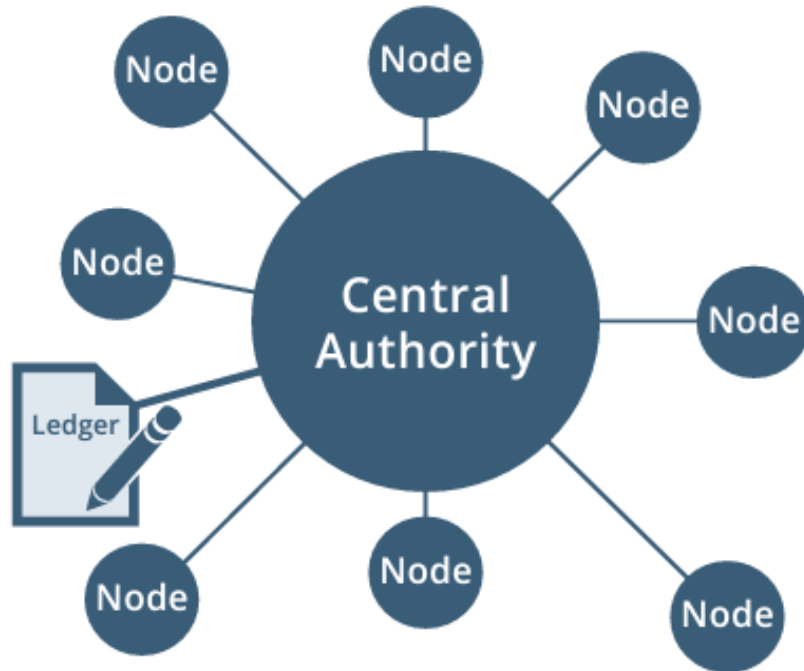
- In autumn 2008, an anonymous person or group known as **Satoshi Nakamoto** published the *Bitcoin Whitepaper*.
- **Bitcoin was launched in 2009;**
- This was the first example of **blockchain technology** applied as a currency system
- Blockchain technology can be summed up as **decentralization through a shared ledger of transactions**
- The three main components are
 1. a **peer-to-peer network** with randomized groups, or nodes;
 2. a database, or **digital ledger**; and
 3. third parties.
- When a third party submits an entry or payment, to the ledger, the nodes work together seamlessly to either **approve or reject transactions**.
- With **no central authority**, this eliminates the need to trust one party such as a payment processor.
- Everything is timestamped and protected by **cryptographic signatures**, or complex algorithms that provide data integrity..



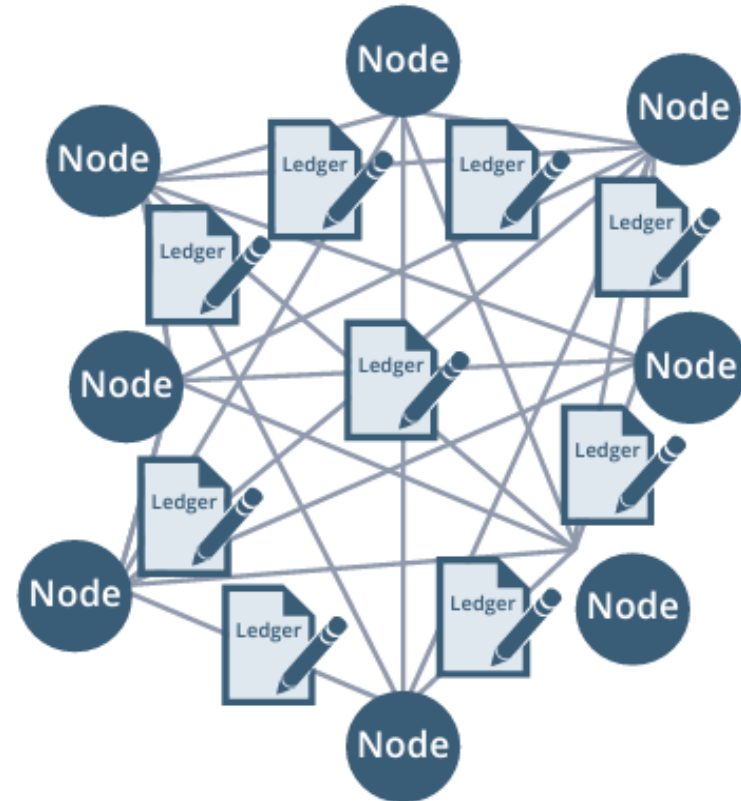
Digital Currencies : Crypto (2)

Removing the central counterparty with decentralized ledger technology

Traditional Ledger



Decentralized Ledger



If any party attempted to retroactively adjust transactions, it would be visible to every node in the network, essentially making transactions fully immutable once submitted.

Digital Currencies : Crypto (3)

- The process of *creating new bitcoin* is by *solving a computational puzzle*.
- This *mining* is necessary to maintain the ledger of transactions upon which Bitcoin is based.
- High-powered computing seeking to solve complex computational mathematical problems, termed '*proof of work*'.
- In addition to producing new Bitcoin, solving each puzzle verifies a block of transaction information, adding them to the public record i.e. the blockchain.
- Nodes within the network then maintain records of those blocks so that they are *verified into the future*.
- The process also ensures *no duplication of a Bitcoin*.
- *Maximum of 21 million Bitcoins*, with mining expected to be *completed around 2140*.
- Huge energy consumption by Bitcoin mining



Digital Currencies : Crypto (4)

Aramonte et al (2021)

Crypto vs traditional financial system¹

Table 1

Function	Service	Crypto financial system		Traditional finance
		Decentralised finance (DeFi)	Centralised finance (CeFi)	
Trading	Funds transfer	DeFi stablecoins (DAI)	CeFi stablecoins (USDT, USDC)	Traditional payment platforms
	Asset trading	Crypto asset DEX (Uniswap)	Crypto CEX (Binance, Coinbase)	Exchanges and OTC brokers
	Derivatives trading	Crypto derivatives DEX (Synthetix, dYdX)		
Lending	Secured lending	Crypto decentralised lending platforms (Aave, Compound)	Crypto centralised lending platforms (BlockFi, Celsius)	Broker-dealers active in repo and securities lending
	Unsecured lending	Crypto credit delegation (Aave)	Crypto banks (Silvergate)	Commercial banks and non-bank lenders
Investing	Investment vehicles	Crypto decentralised portfolios (yearn, Convex)	Crypto funds (Grayscale, Galaxy)	Investment funds

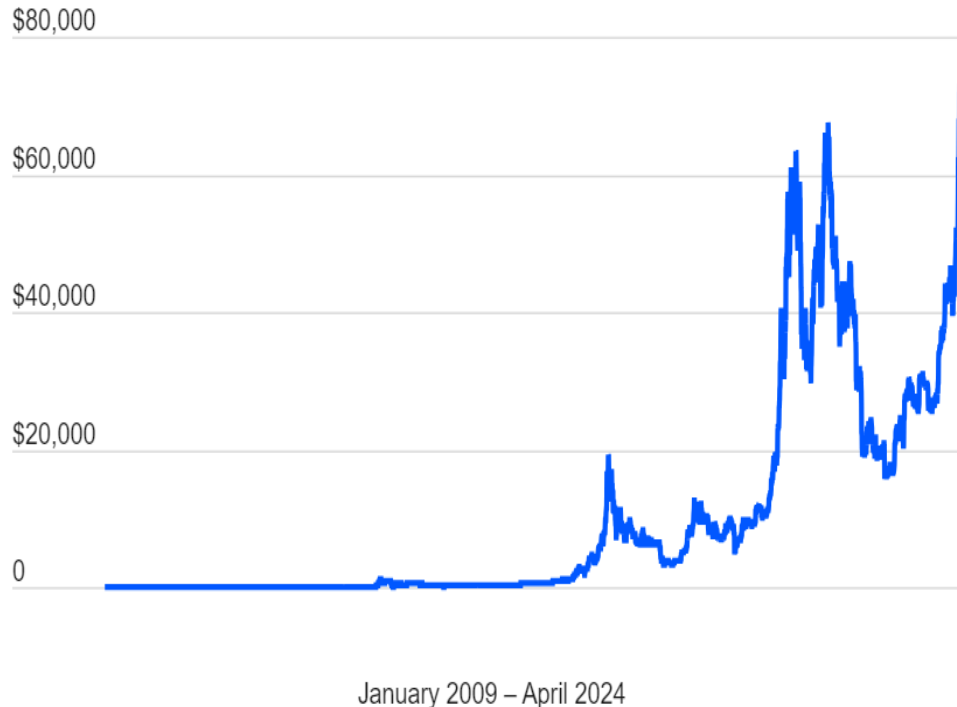
CEX = centralised exchanges; DEX = decentralised exchanges; OTC = over-the-counter; USDC = USD Coin; USDT = Tether.

¹ Illustrative examples are given in parentheses.

Digital Currencies : Crypto : Gains & Losses

Bitcoin's price history since its creation

Bitcoin has risen since its creation in 2009, but not without some bumps along the way.



Source: Historical prices from Investing.com

The Crypto Projects Stealing from Their Investors

Biggest rug pulls by estimated cryptocurrency USD value stolen*



* Rug pull = Cryptocurrency developer abandoning a project/token, draining all funds and disappearing

Sources: Chainalysis, Coinscreed



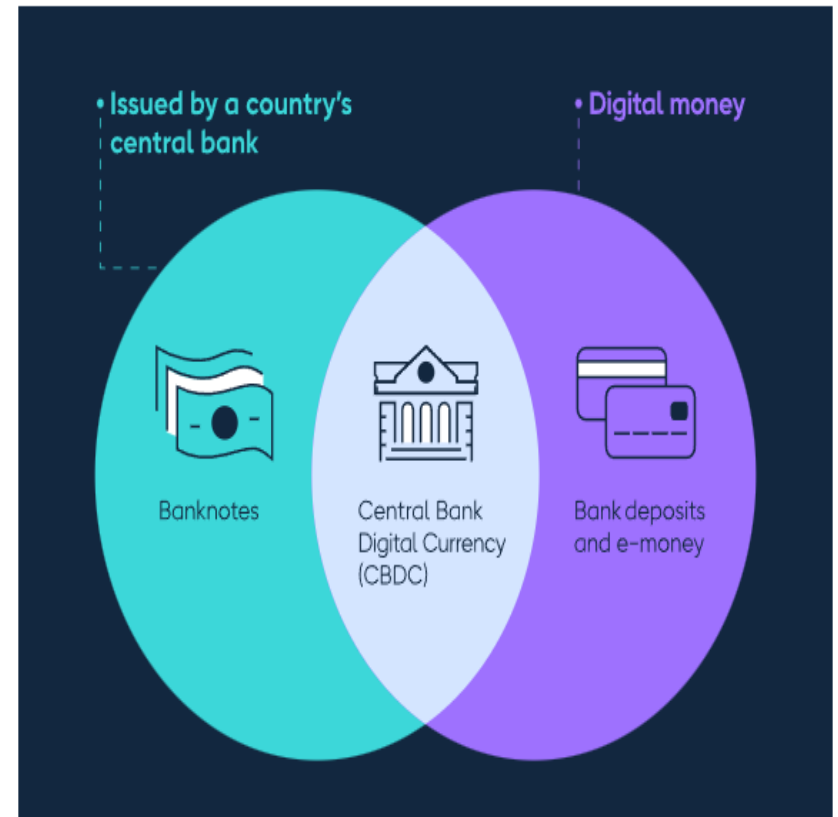
statista

W

Digital Currencies : CBDCs

- **Central bank digital currency (CBDC)** is money that a country's central bank can issue.
- It's called digital (or electronic) because it isn't physical money like notes and coins. It is in the form of an amount on a computer or similar device.
- CBDCs can be classified into “**General Purpose CBDC**” and “**Wholesale CBDC**”.
- General purpose CBDCs are **direct alternatives to cash** for everyday routine use.
- Wholesale CBDCs are used primarily by **payment system operators** based on reserves held with a central bank

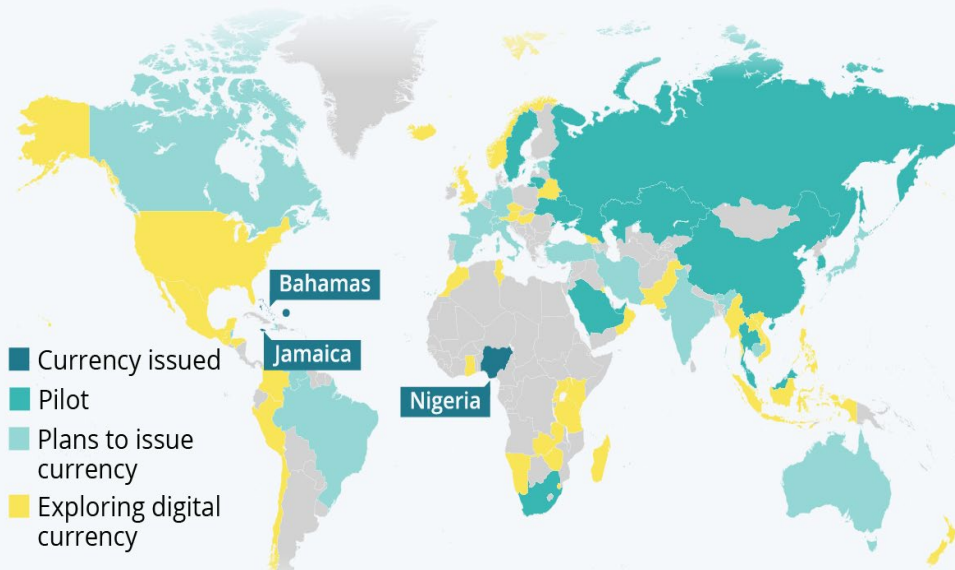
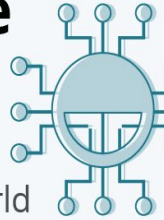
Central bank digital currency is a type of digital money



Digital Currencies : CBDCs (2)

Where Central Banks Have Issued Digital Currencies

Central Bank digital currencies launched, under development or considered around the world



Caribbean nations/territories with official digital currency: Bahamas, Jamaica, Dominica, Grenada, St. Vincent, St. Lucia, Montserrat, St. Kitts, Antigua and Barbuda

Source: Atlantic Council



statista

- **134 countries** & currency unions, representing **98%** of **global GDP**, are exploring a CBDC.
- **3 countries** have fully launched a CBDC—the Bahamas, Jamaica and Nigeria
- 19 of the Group of 20 (G20) countries are now in the advanced stages of CBDC development.
- Brazil, Russia, India, China, and South Africa—the founding members of BRICS—are in the pilot phase of CBDC exploration
- Progress on retail CBDC has **stalled in the US**
- As the **largest CBDC pilot in the world**, China's digital yuan (e-CNY) reaches 260 million wallets across 25 cities.
- Since Russia's invasion of Ukraine and the resulting G7 sanctions response, wholesale **CBDC developments have doubled.**

[Atlantic Council \(2024\)](#)

Fintech Risks and Benefits

Leveraging the digital opportunity does not come without risks



9

9

Fintech Challenges

Top Challenges for Fintech CEOs in the next 12-18 months (selected by % of CEOs)



Top Priorities for Fintech CEOs in the next 12-18 months (selected by % of CEOs)



Sources: BCG/QED Future of Fintech survey (N=81), conducted across fintech CEOs and C-Suite leaders in February 2023; BCG analysis.

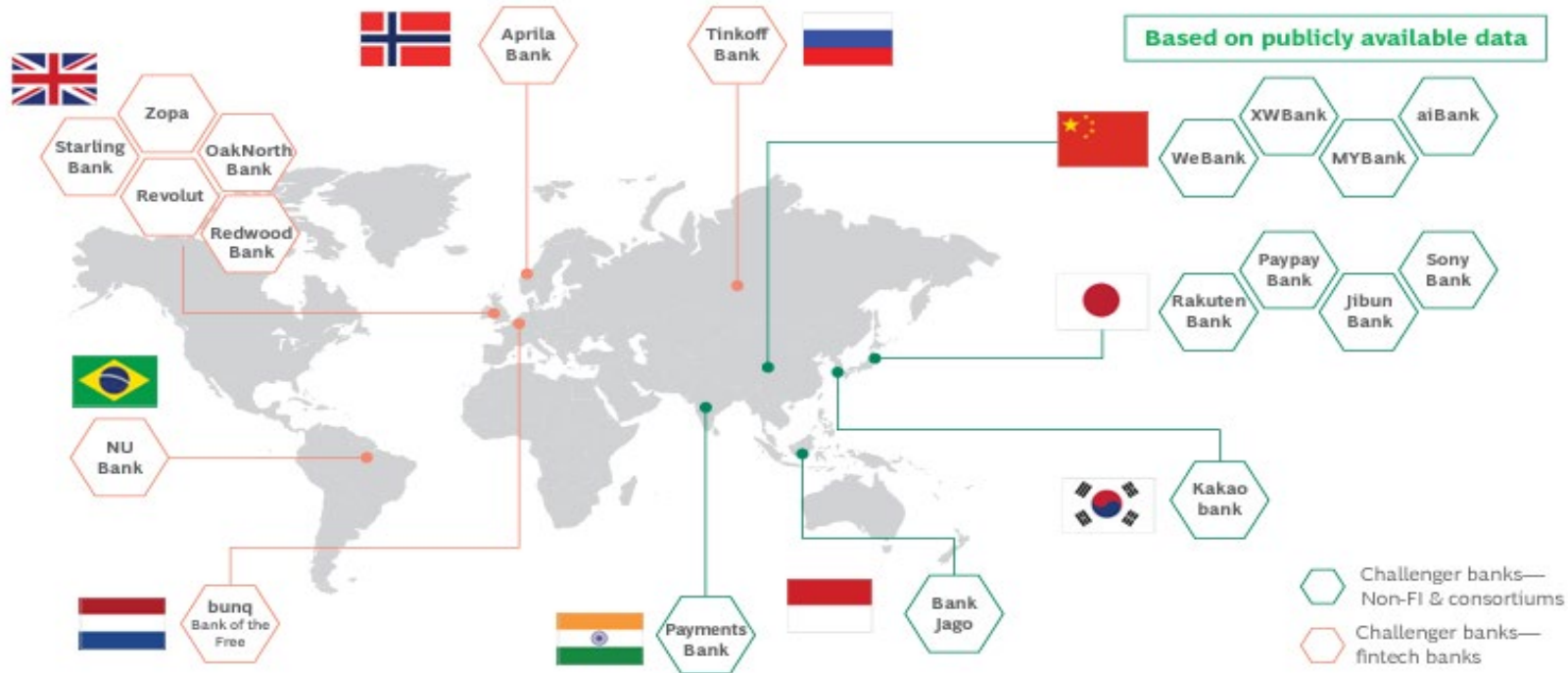
Q. What are the top 3 challenges facing your company in the next 12–18 months?

Q. What are the top actions your company is taking to address challenges over the next 12–18 months?

Note: Responses collected as free-form text and then sorted into discrete categories.

Fintech Challenges (2)

Exhibit 15 - In 2022, Only 20 Out of 453 Digital Challenger Banks Were Profitable, 11 of Which Are in Asia-Pacific



Source: BCG Fintech Control Tower.

Fintech Regulations

- Financial services are the most heavily regulated sectors traditionally
- Fintech companies increase the regulation problems/concerns among governments.
- Regulation Supporting Fintech
 - UK-Application Programming Interface (API)-a set of routines, protocols, and tools for building software applications
<https://www.gov.uk/government/consultations/data-sharing-and-open-data-in-banking-call-for-evidence/call-for-evidence-on-data-sharing-and-open-data-in-banking>
 - EU-Payment Services Directive 2 (PSD II)-safeguard banking data accessing to enable innovation meanwhile preventing data misuse and security breaches.
- Balancing act between regulation and stifling innovation; regulating activity and technology; rules or principles based

Fintech Regulations Tools

WAIT & SEE	TEST & LEARN	REGULATOR Y SANDBOX	WAIVER/ EXEMPTIONS	LETTERS OF NO- OBJECTION	DIFFERENTIA TED REGULATION	REGULATO RY REFORM/LA W
						
Allows innovations to develop with no restrictions •May work well in environments with limited regulatory capacity •Presents significant risk if not controlled later through regulations •Example: China	New innovations tested in a live environment with involvement from the regulators •Case-by-case basis •Requires active involvement by regulator •Difficult to scale up; hard to ensure equal treatment •Examples: Philippines, Tanzania	Virtual Environment where innovators can test their products/service s in a time bound manner. •Greater transparency and replicability • Relevant for markets with good supervisory capacity and high level of active, potentially non-licensed players	Waiver or exemption from a license or restrictive sections of a license • usually codified in law, therefore no need for subjective decisions from the regulator • No special resources for maintenance are required	legal certificate issued by the authority to indicate that it does not object the product or services provided by the fintech entity. Relevant for smaller markets, when the risk profile of the innovations are generally well-understood. Example: Kenya	Usually codified in law, therefore no need for subjective decisions from the regulator No special resources for maintenance are required Example: India: Payments Bank & Trade Receivables Platform license.	Enactment of regulations that support FinTech startups and consumer protection, competition, financial stability and financial inclusion •Modifications to existing regulations •Examples: EU (PSD2, GDPR), Mexico (FinTech Law)



DEVELOPING & OPERATIONALIZING
NATIONAL FINANCIAL INCLUSION STRATEGIES
FOR THE DIGITAL ECONOMY
Washington DC • December 10–12, 2018



WORLD BANK GROUP
Finance, Competitiveness & Innovation

Fintech Regulations : Challenges

- Balancing act between :
 - *regulation and stifling innovation;*
 - *regulating activity and technology;*
 - *rules or principles based*
 - *Timingwhat stage of innovation lifecycle?*
 - *Framework that captures non-financial service providers*
 - *International cooperation vs national interests*



Additional Reading

- [Central Bank of Ireland: Explainer - What is “fintech” and how is it changing financial products?](#)
- [FinTech Wales](#)
- [Bank of England: Fintech](#)

Thank you, very much!

Any questions?



Prifysgol Wreccsam
Wrexham University

References

Aramonte, S., Huang, W., & Schrimpf, A. (2021). *DeFi risks and the decentralisation illusion*. BIS Quarterly Review, 6.

https://www.bis.org/publ/qtrpdf/r_qt2112b.htm?utm_campaign=This%20week%20in%20fintech&utm_medium=email&utm_source=Revue%20newsletter

Bank of England. (2023, January 12). *Fintech | Bank of England*.

<https://www.bankofengland.co.uk/research/fintech>

Panzarino, H and A. Hatami (2021), *Reinventing Banking and Finance: Frameworks to navigate global fintech innovation*, Kogan Page.